



REDE GAZETA



Roadmap

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- Initial Model Prototype
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Project Objective

This project aims to implement a dynamic pricing system to optimize the occupancy of radio advertising slots, adapting in real-time to variables such as demand, seasonality, and peak hours. The initiative seeks to maximize revenue through automatically adjusted prices while offering attractive and predictable conditions for advertisers.



Module summary

- Ongoing communication with the Rede Gazeta team for alignment and data delivery
- Implementation of data governance best practices, including confidentiality, LGPD compliance, and access control
- Execution of exploratory data analysis using tools such as pandas, visualizations, and summarizations
- Creation and automation of a structured ETL pipeline for data ingestion, transformation, and preparation
- Development of the initial cloud architecture template, with a focus on AWS
- Structuring of the data foundation and pipeline for future use in Machine Learning models



Data Governance

Confidentiality Policies.

Restricted access to data is limited to the team only; client and executive names are handled with confidentiality.

Compliance with the LGPD

Removal or masking of sensitive data; minimal use of information necessary for analysis.

Controls and Best Practices

No public upload of data; local control and documentation of the data processing procedure.



Data Processing and Analysis

1. Structure and Diagnostics

Mapping of columns, data types, duplicates, and missing values using pandas

2. Statistical Summary

Use of `.describe()` for numerical and categorical variables

3. Initial Visualization

Charts and tables to detect patterns, outliers, and relevant categories



ETL

1. Extract

Importing files (.csv, .xlsx) via recursive reading of standardized directories.

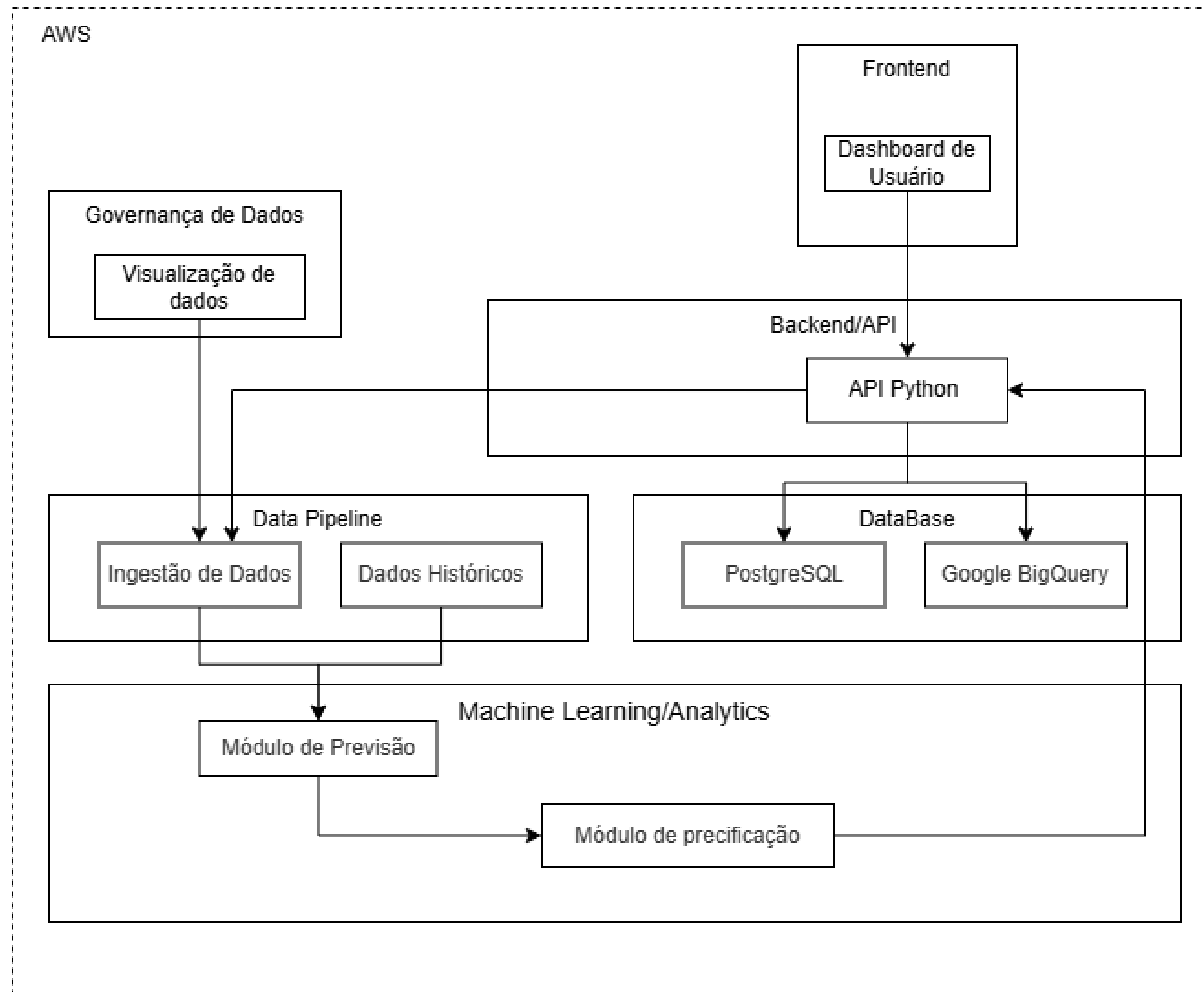
2. Transform

- Data cleaning: dates, values, names.
- Normalization and standardization.

3. Load

Sparse matrix generation and structure ready for dashboards and future models.

Initial Solution Architecture



Initial Model Prototype

- Developed a first version of a pricing model
- Aims to determine the optimal price to maximize ad slot occupancy
- Serves as a foundation for the future dynamic pricing engine



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Next Module – Planning and Focus

1. Model Development

- Build the dynamic pricing engine.
- Choose and validate machine learning algorithms

2. Technical Integration

- Develop API to serve the model.
- Begin planning the front-end interface.

3. Infrastructure and Operationalization

- Run automated data pipeline on the cloud (AWS).
- Monitoring, versioning, and initial real-world tests.





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